

Non-submission research record: forensic audit of interaction-structure claims in a static wake model

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Abstract

This non-submission record supersedes an earlier draft that claimed a complement-substitute decomposition, a decoupling law, and a greedy certificate for yaw optimization. A falsification-oriented audit found that its mathematical scope and numerical evidence did not support those claims. The derivation assumed separable deficit kernels and recovery monotonicity, whereas the FLORIS Gauss-curl hybrid (GCH) model used for numerical illustrations includes yaw-added recovery and secondary steering. In a reproducible two-turbine, laterally offset GCH case, increasing positive upstream yaw from 0 to 5 degrees lowers downstream power by 46.175 kW; recovery monotonicity is therefore not automatic for arbitrary geometry. At the former headline three-turbine point, the mixed finite difference changes from minus 0.215420 kW deg⁻² at a 5-degree step to plus 0.022315 kW deg⁻² at a 1-degree step, so the reported phase flip is not a verified local-Hessian result. In addition, sampled derivatives do not establish the box supremum needed by the former interaction-energy bound. The document records the falsifications, withdraws the affected claims, and specifies requirements for any future research. It does not report a new physical law, controller, or submission-ready result.

1 Introduction

This document is an archival correction, not a research manuscript for peer review. It replaces an earlier P1 draft whose title and abstract suggested that a broadly applicable interaction structure had been established for wake-steering objectives. The earlier draft must not be submitted or cited as a result. The reproducible source, structured output, and fuller decision record are retained with this research record; their exact locations are listed in the data-availability statement.

The distinction matters because a conditional derivative identity for an idealized map is not automatically a property of an engineering wake model. GCH was designed precisely to add yaw-induced recovery and secondary steering to a Gaussian wake description (King et al., 2021). Those effects are useful for wake-steering studies, but they introduce dependencies that the earlier separable-kernel derivation did not model. Optimizer sensitivity and nonsmooth or discontinuous behavior in engineering wake objectives are also documented in the literature (Gori et al., 2023). The old text incorrectly treated these issues as harmless changes of magnitude.

2 Claims withdrawn

Model-class statement. The former derivation took a receiver velocity to be a sum or sum-of-squares combination of kernels $w_{ij}(\gamma_i)$ and assumed $-\partial w_{ij}/\partial \gamma_i \geq 0$. That can define a useful conditional mathematical model, but it does not contain GCH as a special case. In GCH, upstream yaw can alter the effective yaw and wake behavior of a downstream turbine through secondary steering (King et al., 2021). The prior claims that all GCH variants satisfy the decomposition and that secondary steering cannot affect its sign structure are withdrawn.

There was also a formal presentation error. A mixed partial is symmetric, while the old direct-substitution term was written for arbitrary $i \neq j$ using only the indicator $\mathbf{1}\{j \succ i\}$. Any future proposition must either declare an ordered upstream–downstream pair or include both possible directed terms. It must state that analytic yaw derivatives use radians; the old finite-difference diagnostics were reported in kW deg^{-2} .

Numerical phase claim. The former complement-to-substitute phase flip in a three-turbine chain was inferred from a central difference with a 5-degree step. That number is not stable under local refinement, as shown in Table 1. It is consequently inappropriate to call it a local Hessian sign, phase boundary, or validation of a continuous derivative formula.

Table 1: Falsification check at the former three-turbine-chain state $(20^\circ, 20^\circ, 20^\circ)$. Values are central finite-difference diagnostics for the first two yaw coordinates under one FLORIS 4.6.6 GCH condition. They are not derivative bounds.

Difference step	Dagnostic M_{12} (kW deg^{-2})
5°	-0.215420
2.5°	-0.248412
1°	$+0.022315$
0.5°	$+0.022367$
0.25°	$+0.022381$

Decoupling and greedy claims. A selected finite set of Hessian ratios cannot establish a law of decoupling at optima. Likewise, the former interaction-energy expression required a supremum over a yaw box, but the numerical study evaluated a handful of finite-difference samples. Sampling can form a heuristic numerical screen; it cannot certify a global derivative envelope, a greedy optimality gap, a clustering loss, or a contraction factor. The words *theorem*, *law*, *guarantee*, *certificate*, and *provably safe* are therefore withdrawn for the former P1 results.

3 Reproducible counterexample to an automatic premise

The audit uses FLORIS 4.6.6 with the NREL-5MW default input. Inflow is 8 m s^{-1} , turbulence intensity is 0.06, wind direction is 270 degrees, and the yaw box is one-sided from 0 to 30

degrees. Two turbines are 5 rotor diameters apart in the streamwise direction; the receiver is offset laterally by $-1D$ and held at zero yaw. Table 2 reports its modeled power as the upstream yaw increases.

Table 2: A laterally offset FLORIS GCH case in which positive upstream yaw moves the wake toward the receiver. This numerical counterexample is scoped to the stated model and geometry; it does not make a claim about all physical wind farms.

Upstream yaw	Downstream modeled power (kW)
0°	1651.808
1°	1643.087
5°	1605.633

The 0-to-5-degree change is -46.175 kW. Thus a positive yaw direction does not by itself establish the recovery-monotonicity premise in an arbitrary layout. This behavior is consistent with the importance of correct versus wrong-way steering in field-model comparisons (Fleming et al., 2021). It invalidates the former universal/model-family wording, rather than establishing a new empirical law.

4 Experimental and publishing boundary

No wind-tunnel, LES, or field experiment was conducted for the old interaction claims. A future experimental outline is neither a completed experiment nor a preregistration; it must not be described as either. Existing wake-steering experiments are valuable context, but they do not validate the withdrawn mixed-partial conclusions. For example, fixed-angle field tests were designed to evaluate model behavior under both beneficial and detrimental steering conditions (Fleming et al., 2021).

The former draft received substantive assistance from a generative AI system in this workflow. Copernicus policy reviewed on 2026-08-31 prohibits using generative AI to create manuscript text or scientific explanations. Neither metadata edits nor an author declaration can convert this record into a WES submission. The named author would need to independently reconstruct, check, and write any future manuscript under the then-current policy.

5 Requirements before reopening the topic

A future P1-like study would need all of the following before it could be evaluated as a research-paper candidate:

- a correctly stated and independently checked theorem for an explicit model, including angle units, pair ordering, differentiability, and the exact conditions that imply any sign result;

- either analytic derivatives, automatic differentiation with demonstrated regularity, or validated numerical enclosures before any derivative-sign or global-bound claim;
- predeclared tests across layouts, yaw signs, inflows, uncertainty, model classes, and finite-difference refinement, with counterexamples retained;
- a clear separation between a conditional model proposition and behavior observed in GCH, LES, wind-tunnel, or field data; and
- a fresh multi-channel novelty audit after a distinct and validated contribution exists.

6 Conclusions

The earlier P1 claims are withdrawn because their scope, numerical stability, and evidence level did not support them. The retained materials make the failure reproducible: an automatic recovery-monotonicity premise fails in an allowed FLORIS geometry, a central mixed difference reverses sign under refinement at the old headline state, and finite samples do not create a global certificate. This is a non-submission forensic record, not evidence for a new interaction law or an optimizer.

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References

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